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Booking Patterns for Cancellation Risk Management

Unsupervised Learning

Project - Deliverable 1 - Pitch

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Turno: TP1

1.Problem Statement

This project conducts a reproducible clustering study on hotel booking records to identify distinct pre-arrival booking profiles based on behavioral data available strictly at the moment of reservation. We are particularly interested in the dynamics of bookings made far in advance through indirect channels. Because cancellations represent a massive source of lost revenue, identifying if these specific structural profiles are naturally distinct and inherently prone to cancellation is critical for risk management. By discovering unbiased, natural groupings from a broad set of behavioral features, we aim to confirm if this specific profile naturally emerges. If post-hoc analysis reveals varying cancellation rates across this (or other) specific clusters, management can strategically implement targeted countermeasures, such as adjusted overbooking limits or deposit policies, rather than applying inefficient blanket policies.

2.Problem Framing (RQ1)

RQ1: "Do bookings made far in advance through indirect channels naturally emerge as a distinct pre-arrival cluster, and how does this profile associate with observed cancellation rates?"

We aim to segment individual booking records using characteristics available strictly at our defined index time: immediately after the initial booking is recorded. While our primary hypothesis focuses on early-planning, indirect bookings, we must use a broad set of pre-arrival characteristics (stay design, party size, channel, timing) to ensure we do not force this cluster into existence by construction. A useful segmentation will reveal the hidden behavioral structures of customer acquisition, confirming if the "far-in-advance/indirect" profile naturally separates itself from the rest of the demand. Identifying these structural groups allows us to use post-hoc profiling to see which profiles actually associate with different cancellation rates, directly supporting accurate inventory decisions.

Clustering is appropriate because this is an exploratory segmentation task. To test our hypothesis without bias, we cannot rely on predefined segment labels. We must rely on strict feature governance to prevent information leakage, allowing an unsupervised approach to group the data based on inherent similarities, thereby proving whether the early/indirect profile is a genuine, mathematically distinct group.

3.Representation / Preprocessing Choices (RQ2)

Our representation is designed to capture a holistic view of booking behavior at the index time. As noted in methodological feedback, we must avoid building our hypothesis into the representation. Therefore, our baseline representation will be intentionally broad to allow the "far-in-advance/indirect" cluster to emerge naturally, rather than by construction.

The core variables in the baseline representation will include ***lead_time***, ***distribution_channel***, ***market_segment***, ***deposit_type***, ***hotel***, ***customer_type***, stay length (***stays_in_weekend_nights***, ***stays_in_week_nights***), party composition

(*adults*, *children*, *babies*), and prior reliability (*is_repeated_guest*, *previous_cancellations*, *previous_bookings_not_canceled*). By including stay and party dynamics, we ensure bookings are grouped by overall behavior, not just channel and timing. As suggested, we may later introduce a controlled, focused variant that isolates timing and channel variables to explicitly test the robustness of our target profile.

We will explicitly exclude *is_canceled*, *reservation_status*, and *reservation_status_date* from clustering because they are outcome variables reserved exclusively for post-hoc profiling. We will also exclude *agent* and *company*, since they are ID-like and high-cardinality, as well as operational variables like *assigned_room_type* and *booking_changes* which occur after the index time.

Preprocessing & Evaluation Protocol: Numerical variables will be standardized using **StandardScaler**. Categorical variables will have rare levels grouped and be full one-hot encoded without variance standardization. To ensure robust results, we will explicitly test for feature dominance by running controlled comparisons to verify that categorical variables like *distribution_channel* or *market_segment* do not artificially dominate the Euclidean distance space. Our evaluation protocol is strictly defined: We will compare a K-means baseline and iK-means on a bounded range of K against an alternative clustering family, computing internal validity indices consistently in the exact same representation space, and validating stability across 5 random seeds using ARI.

4. Limitation, Risk, or Ethical Issues

Because our baseline representation relies on one-hot encoded categorical variables, such as *market_segment*, *distribution_channel*, and *customer_type*, it may produce a sparse high-dimensional space in which Euclidean distances are less informative. This does not invalidate the required baseline, but it is an important limitation to monitor when interpreting k-means results and when comparing them with alternative clustering approaches or preprocessing variants.

During the post-hoc profiling of our clusters, we will strictly adhere to ethical data practices to ensure our business recommendations remain objective. Specifically, we must be careful not to conflate "rare" or atypical booking patterns with "bad" or undesirable behavior. Furthermore, because our goal involves evaluating cancellation risks to potentially recommend stricter booking policies, we must exercise extreme caution to prevent the overgeneralization of customer traits based on demographic variables, such as nationality (country). Recommending penalizing policies based on biased demographic clusters could lead to unfair or discriminatory business practices. By acknowledging this, we ensure our resulting risk-mitigation strategies are driven by actual, unbiased booking behaviors rather than demographic assumptions.